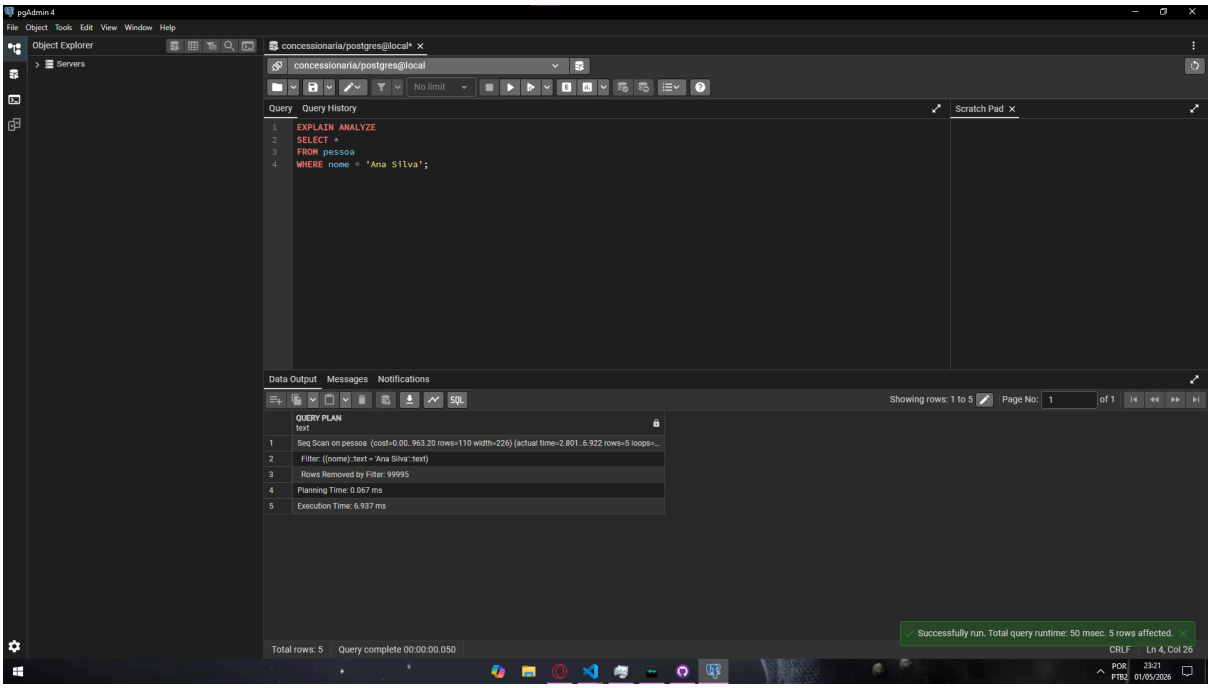
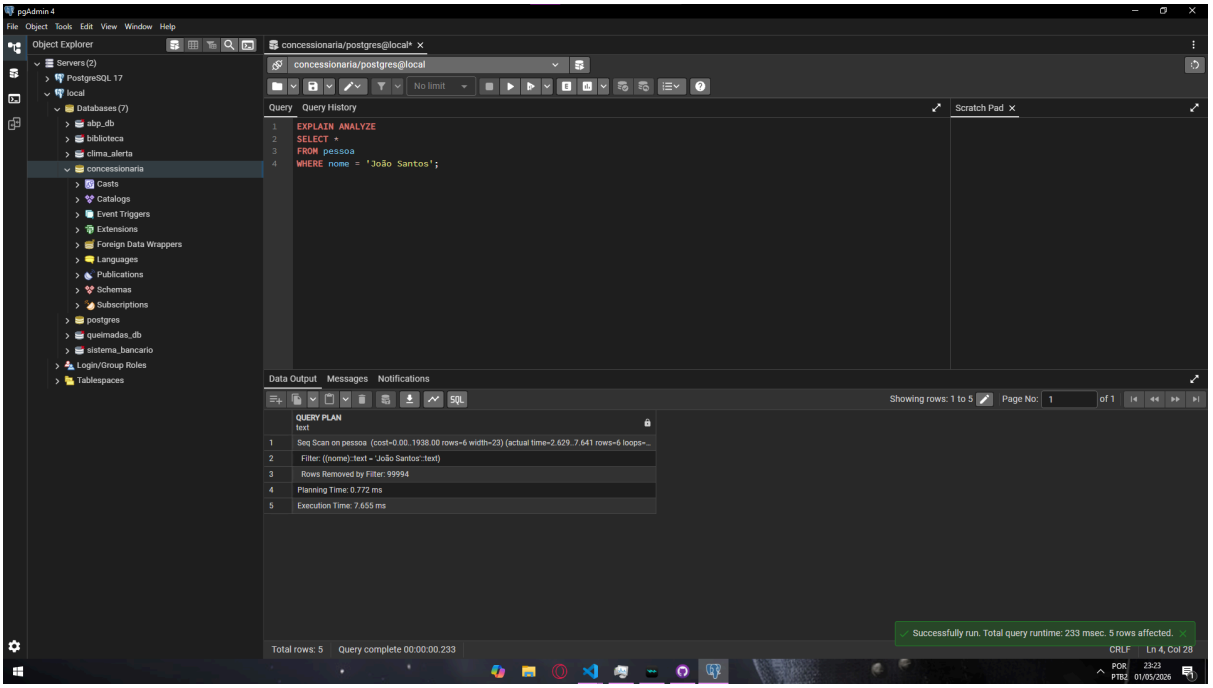
Pedro Augusto Gomes

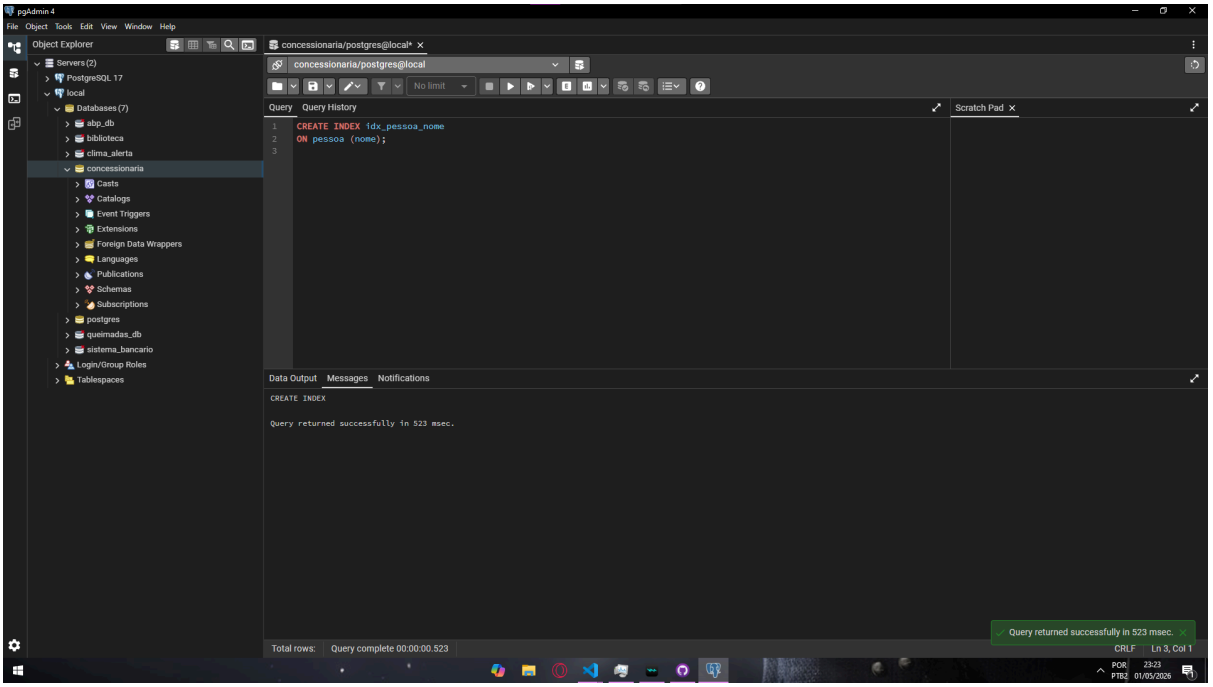
1.

A)





B)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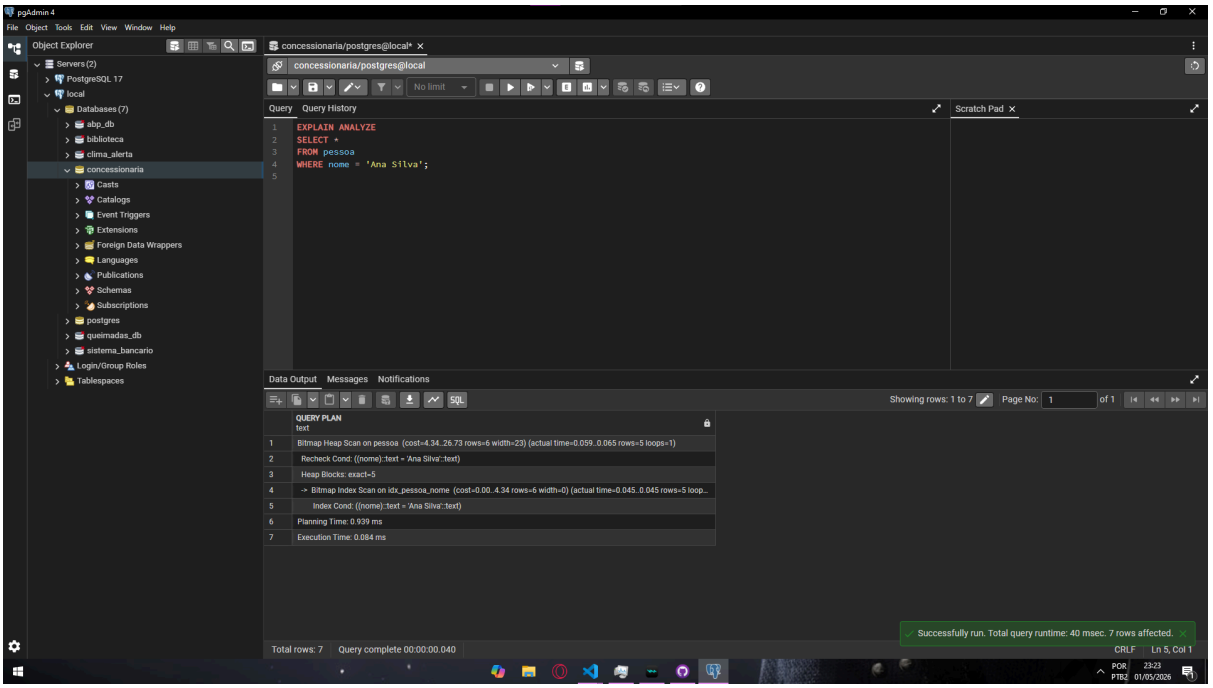
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
CREATE INDEX idx_pessoa_nome
ON pessoa (nome);
Data Output Messages Notifications
CREATE INDEX
Query returned successfully in 523 msec.
Total rows: Query complete 00:00:00.523
Query returned successfully in 523 msec.
CRLF Ln 3, Col 1

C)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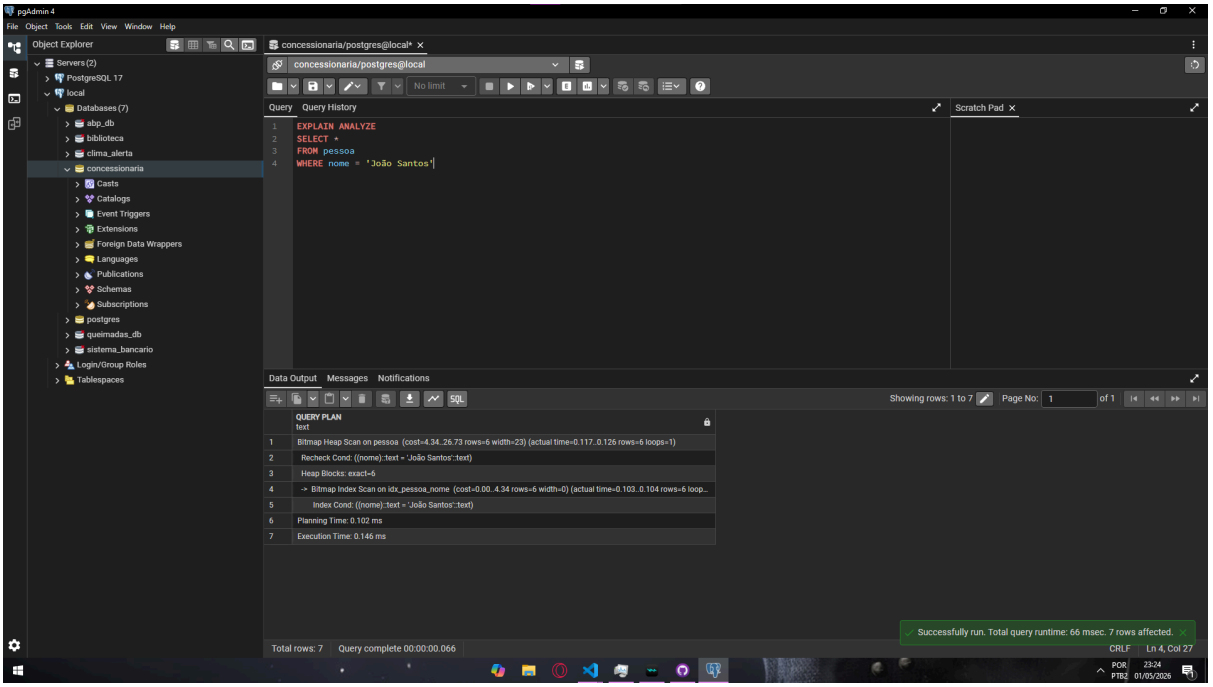
sistema_bancario
Login/Group Roles
Tablespaces
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
EXPLAIN ANALYZE
SELECT *
FROM pessoa
WHERE nome = 'Ana Silva';
Data Output Messages Notifications
SQL
Showing rows: 1 to 7
Page No: 1 of 1
QUERY PLAN
text
1 Bitmap Heap Scan on pessoa (cost=4.34..26.73 rows=6 width=23) (actual time=0.059..0.065 rows=5 loops=1)
2 Recheck Cond: ((nome)::text = 'Ana Silva'::text)
3 Heap Blocks: exact=5
4 -> Bitmap Index Scan on idx_pessoa_nome (cost=0.00..4.34 rows=6 width=0) (actual time=0.045..0.045 rows=5 loop...
5 Index Cond: ((nome)::text = 'Ana Silva'::text)
6 Planning Time: 0.939 ms
7 Execution Time: 0.084 ms
Total rows: 7 Query complete 00:00:00.040
Successfully run. Total query runtime: 40 msec. 7 rows affected.
CRLF Ln 5, Col 1



D)

• Qual plano foi utilizado antes do índice?

Seq scan

• Qual plano foi utilizado após o índice?

Bitmap Heap Scan

• Houve redução no tempo de execução?

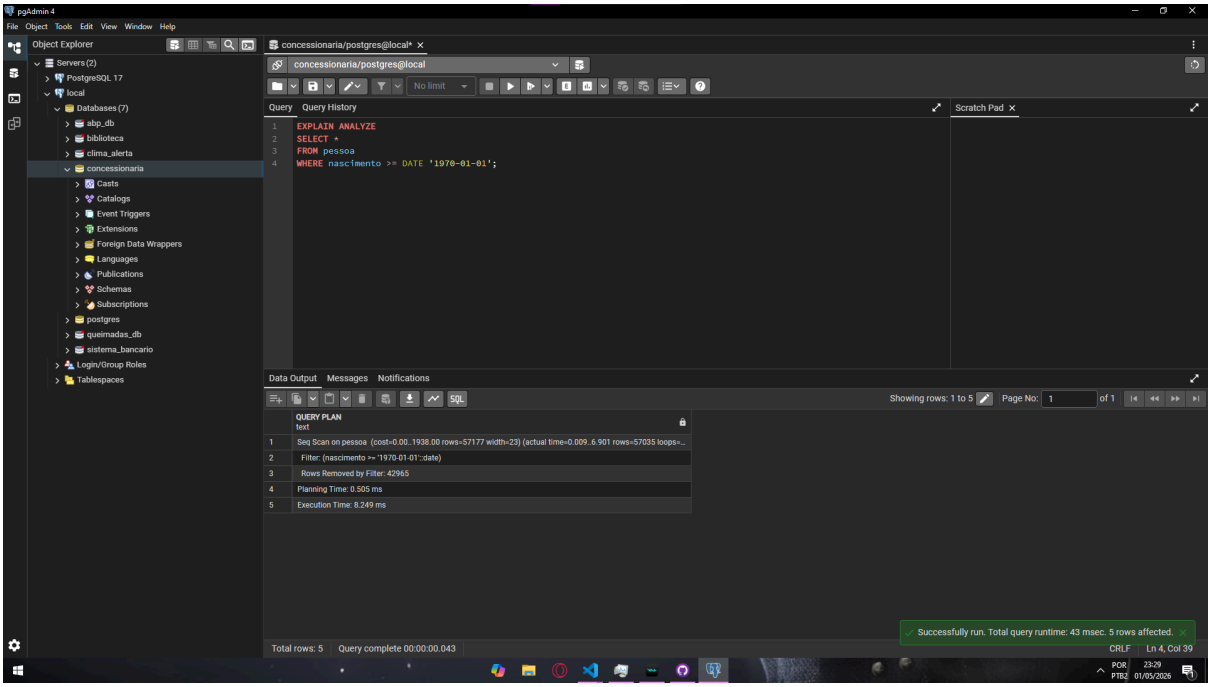
Sim, de aproximadamente 7ms

• O índice foi utilizado em ambos os nomes testados? Justifique com base na saída do plano

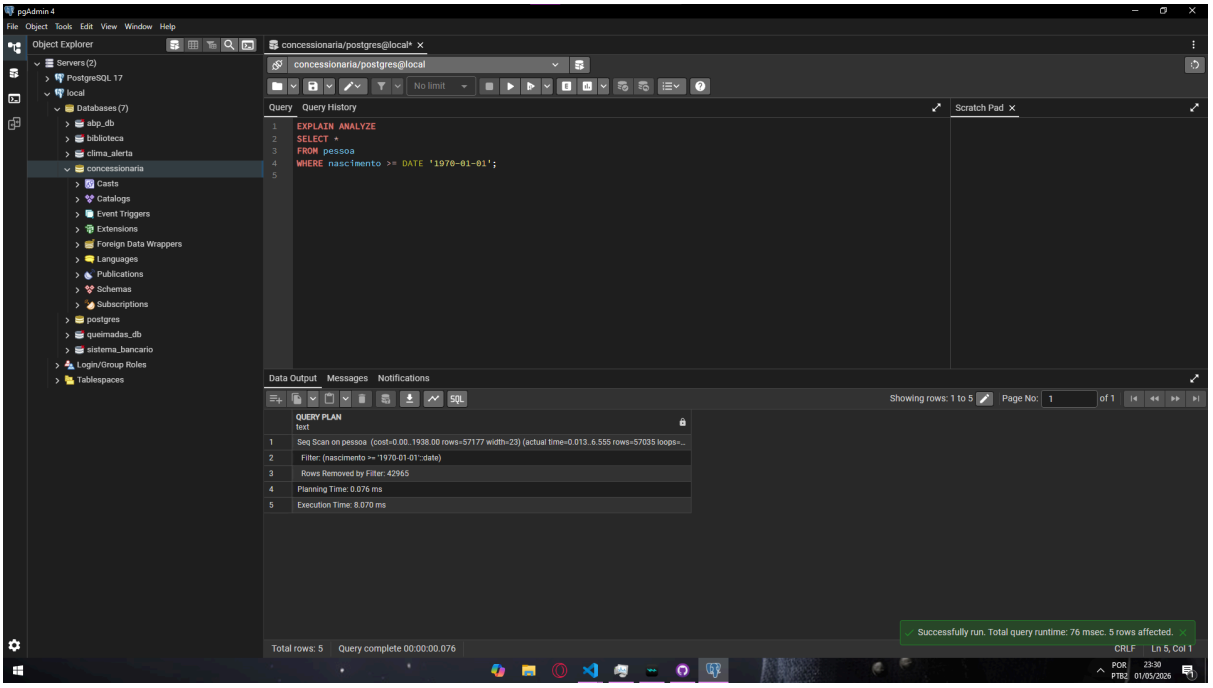
Sim. Como pode ser visto na linha 4 e 5 das últimas imagens, é retornado que o idx_pessoa_nome foi utilizado no select.

2.

A)



C)



D)

- **O índice criado foi utilizado pelo PostgreSQL?**

Não

- **O plano de execução mudou após a criação do índice?**

Não

- **Houve melhora significativa no tempo de execução?**

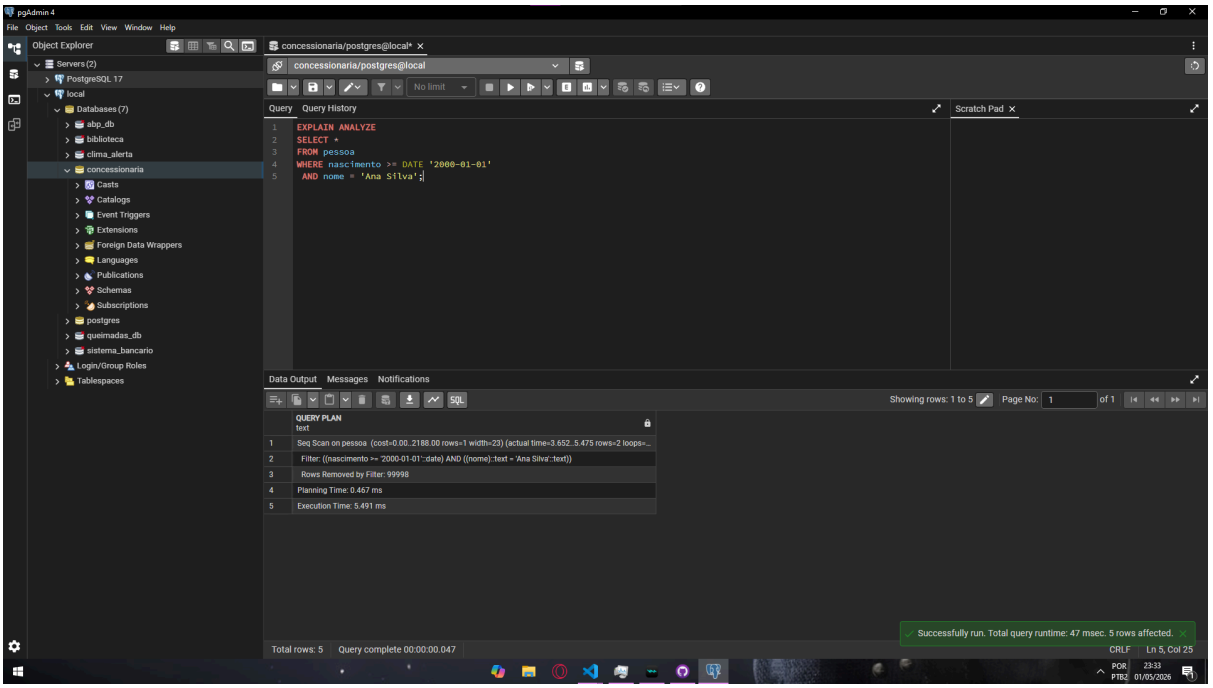
Não, apenas uma pequena melhora.

- **Por que o otimizador pode optar por não utilizar um índice, mesmo quando ele existe?**

Devido a estimativa de custo, tornando o seq scan mais eficiente.

3.

A)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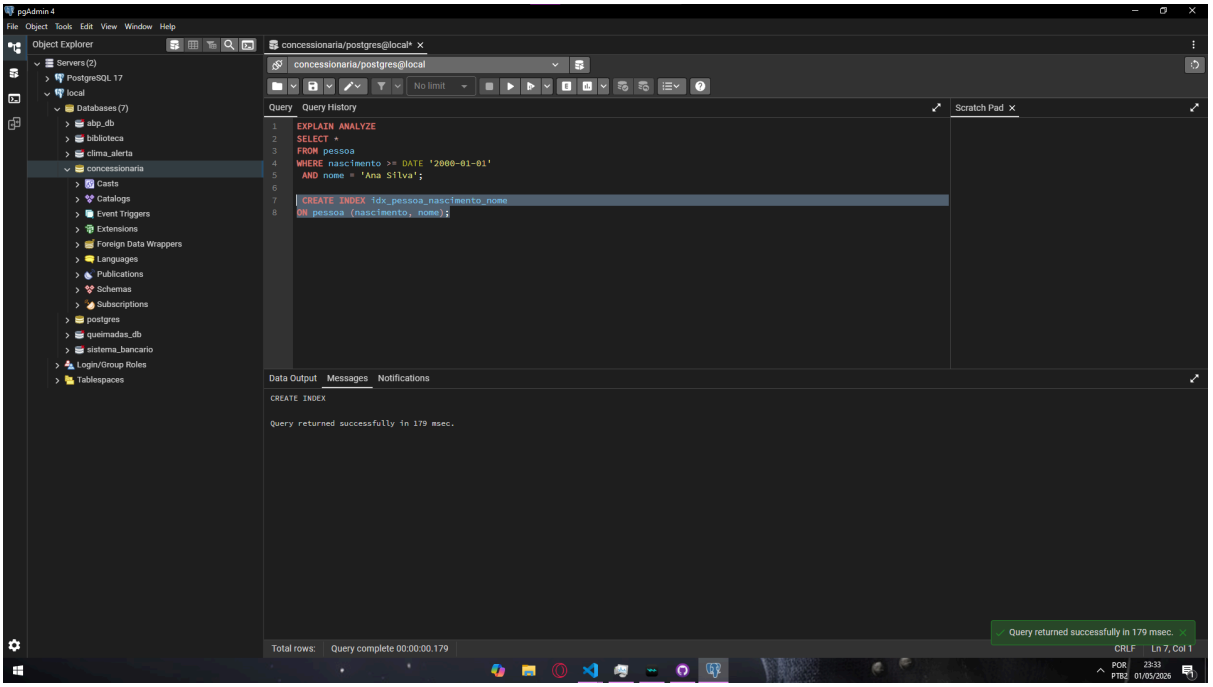
Tablespaces
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
EXPLAIN ANALYZE
SELECT *
FROM pessoa
WHERE nascimento >= DATE '2000-01-01'
AND nome = 'Ana Silva';
Data Output Messages Notifications
Showing rows: 1 to 5
Page No: 1 of 1
QUERY PLAN
text
1 Seq Scan on pessoa (cost=0.00..2188.00 rows=1 width=23) (actual time=3.652..5.475 rows=2 loops=...
2 Filter: ((nascimento >= '2000-01-01'::date) AND ((nome)::text = 'Ana Silva'::text))
3 Rows Removed by Filter: 99998
4 Planning Time: 0.467 ms
5 Execution Time: 5.491 ms
Successfully run. Total query runtime: 47 msec. 5 rows affected.
Total rows: 5
Query complete 00:00:00.047
CRLF
Ln 5, Col 25

B)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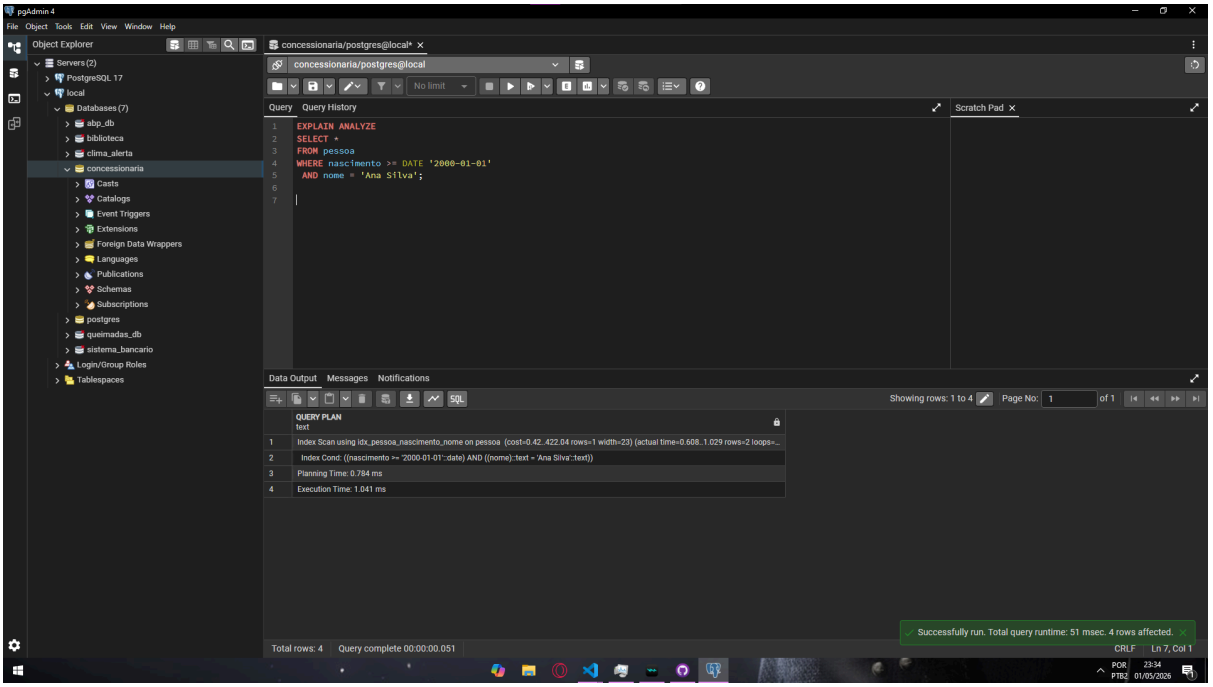
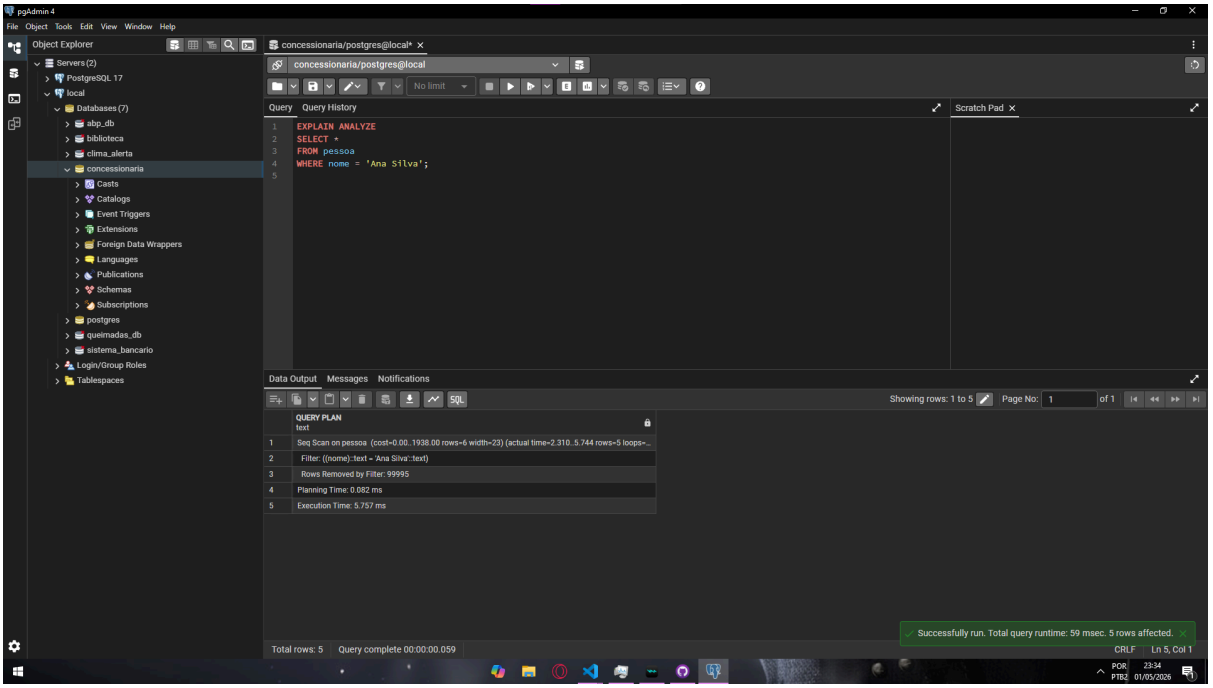
sistema_bancario
Login/Group Roles
Tablespaces
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
EXPLAIN ANALYZE
SELECT *
FROM pessoa
WHERE nascimento >= DATE '2000-01-01'
AND nome = 'Ana Silva';
CREATE INDEX idx_pessoa_nascimento_nome
ON pessoa (nascimento, nome);
Data Output Messages Notifications
CREATE INDEX
Query returned successfully in 179 msec.
Query returned successfully in 179 msec.
Total rows:
Query complete 00:00:00.179
CRLF
Ln 7, Col 1

C)



D)



E)

- **Qual plano foi utilizado antes da criação do índice composto?**

Seq Scan.

- **Qual plano foi utilizado após a criação do índice composto?**

Index scan

- **O índice composto foi utilizado na consulta que filtra apenas por nome?**

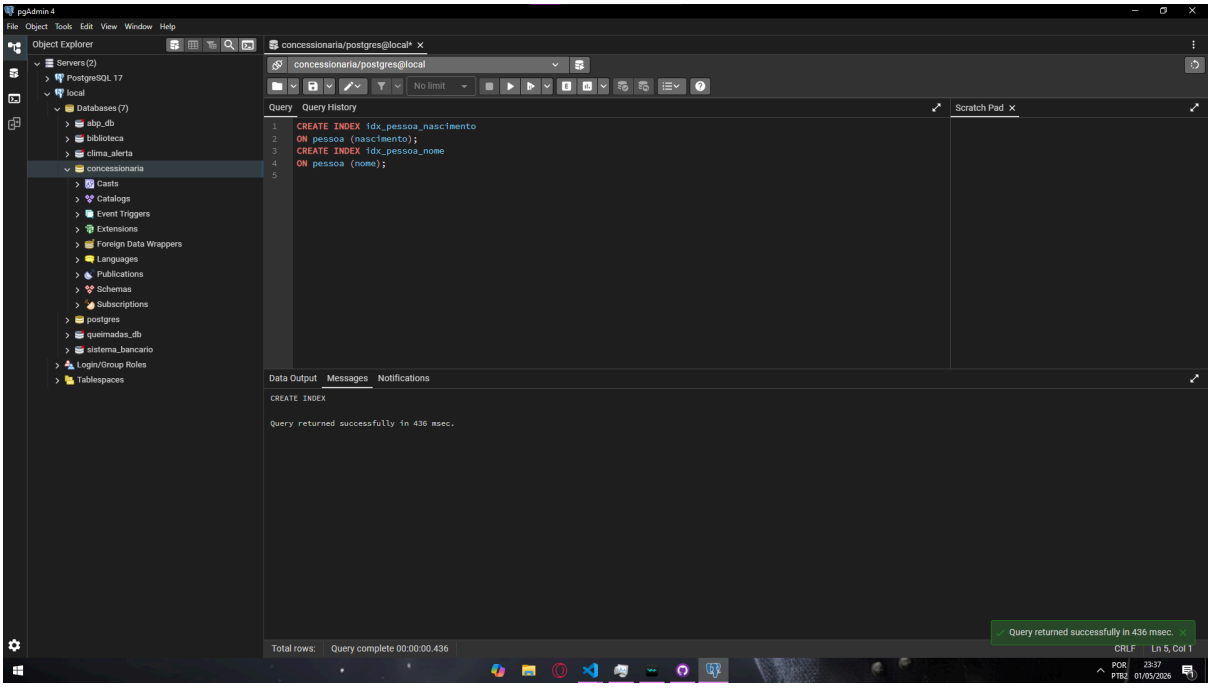
Não

- **Por que a ordem das colunas no índice composto é relevante?**

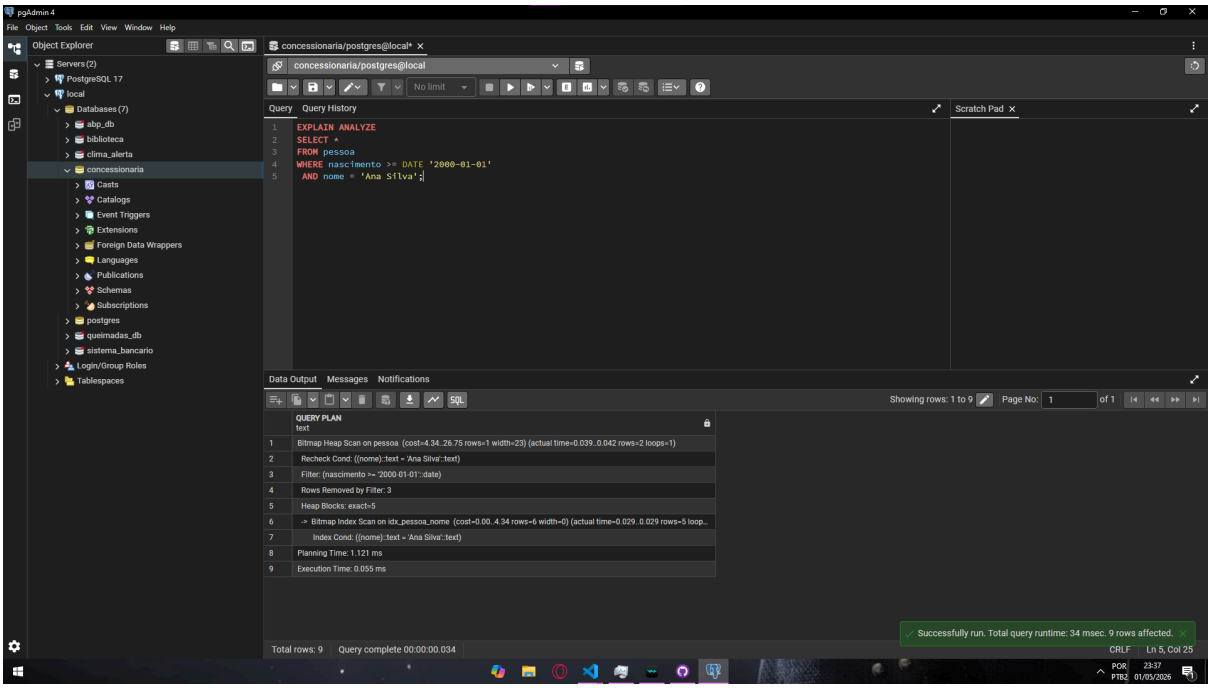
Porque o índice só é eficiente quando a consulta utiliza a primeira coluna definida no index, pessoa (nascimento, nome) nesse caso ‘nascimento’.

4.

A)



B)



C)

- **O PostgreSQL utilizou os dois índices simples na consulta?**

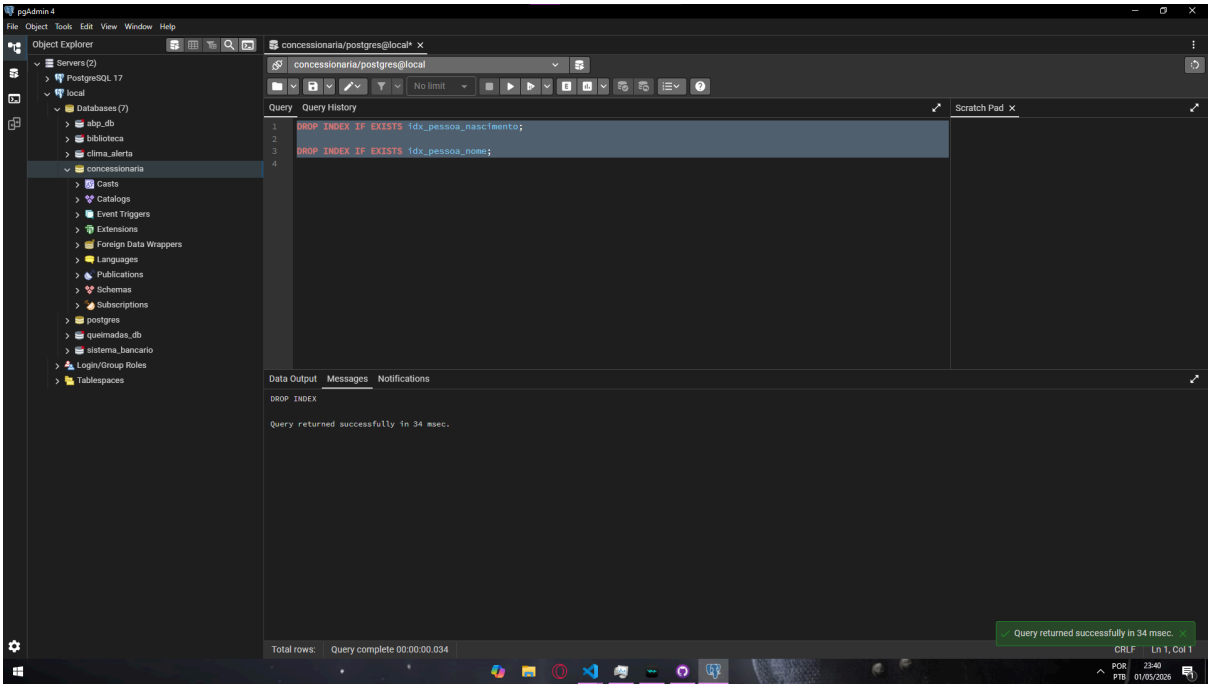
Sim

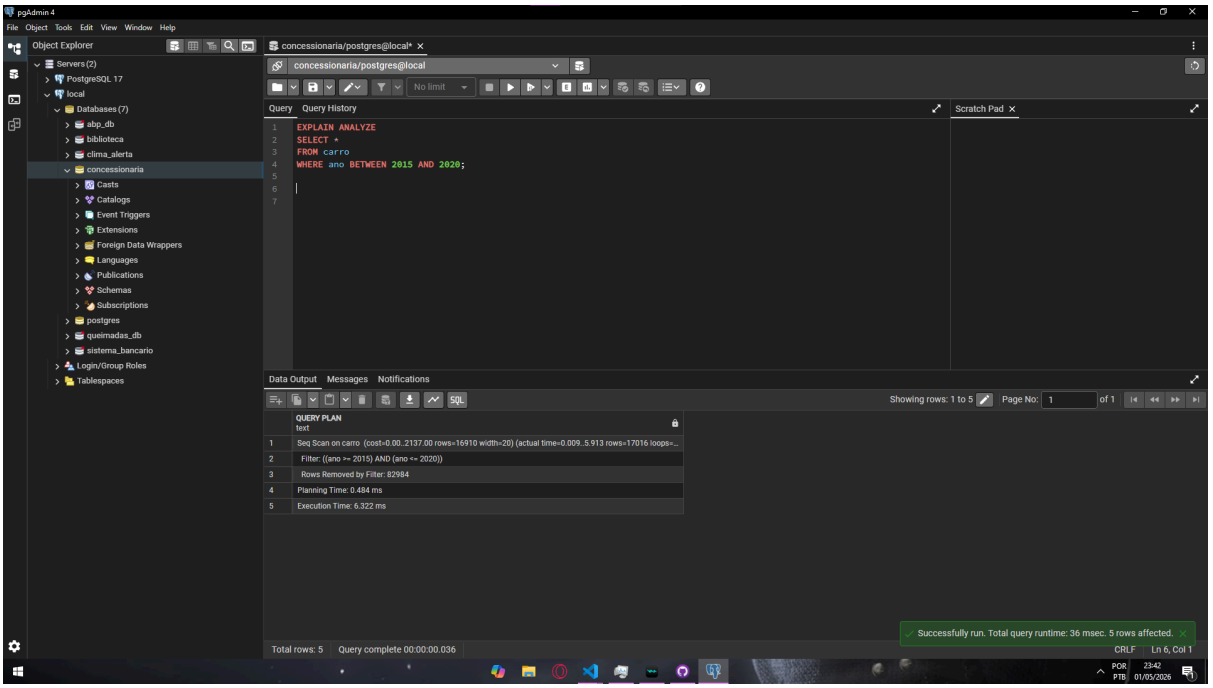
- **Qual é o papel do operador BitmapAnd no plano de execução?**

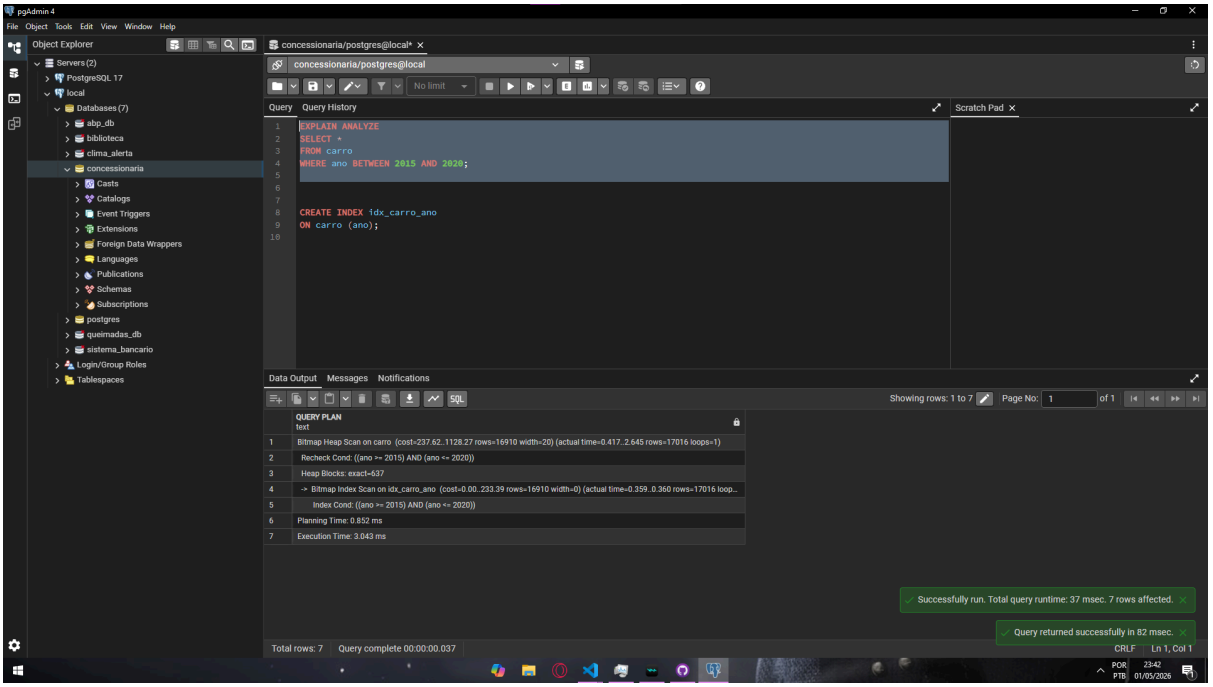
Combinar múltiplos index.

• O que o plano efetivamente fez?
Fez leitura dos dois índices separadamente e combinou os resultados.

5.



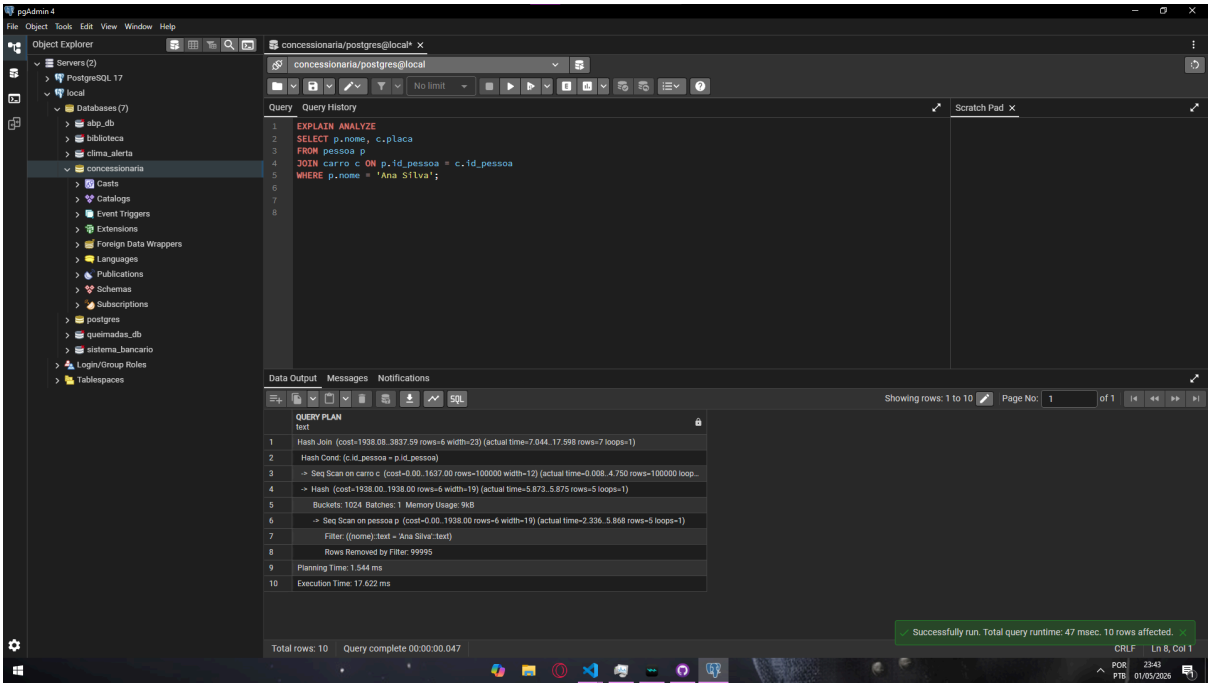


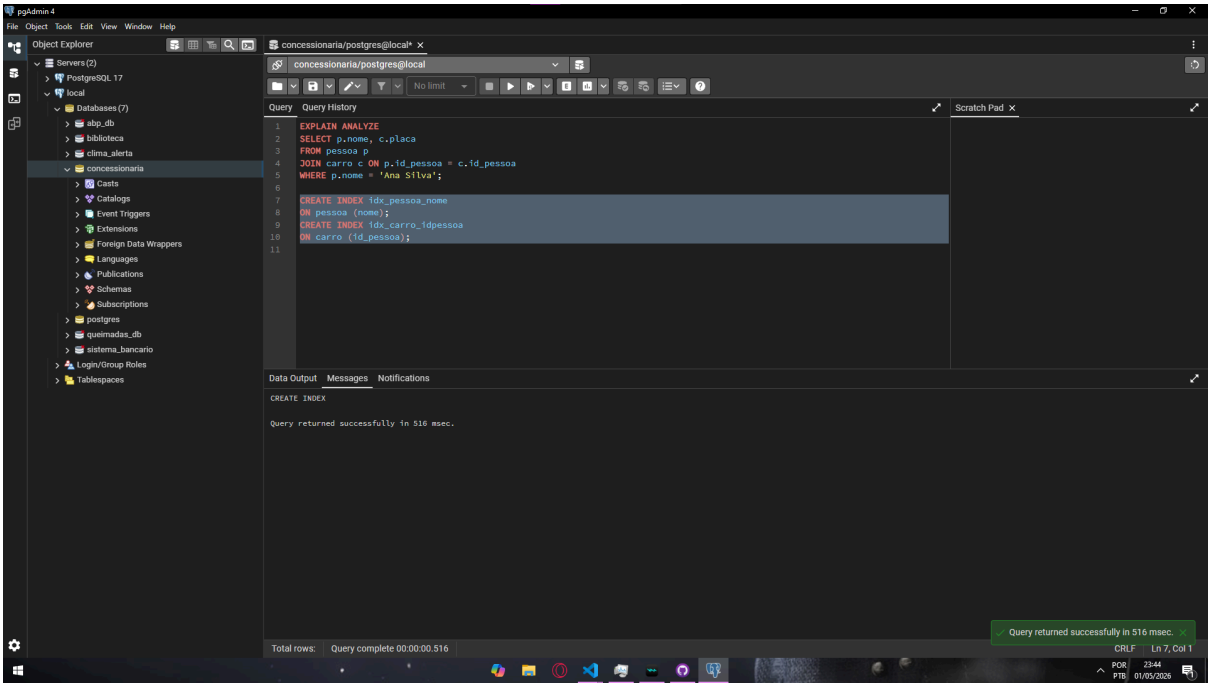


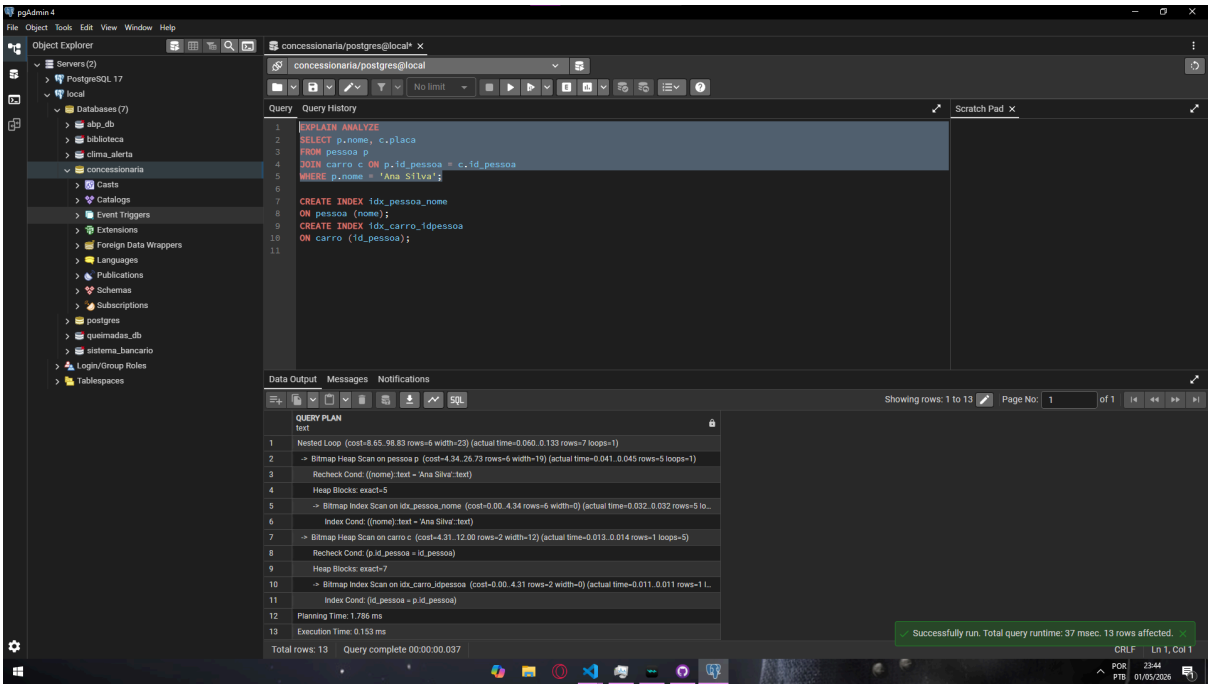
Sem índex: (mais lento)
Seq Scan

Com index: (mais rápido)
Bitmap Index Scan

6.



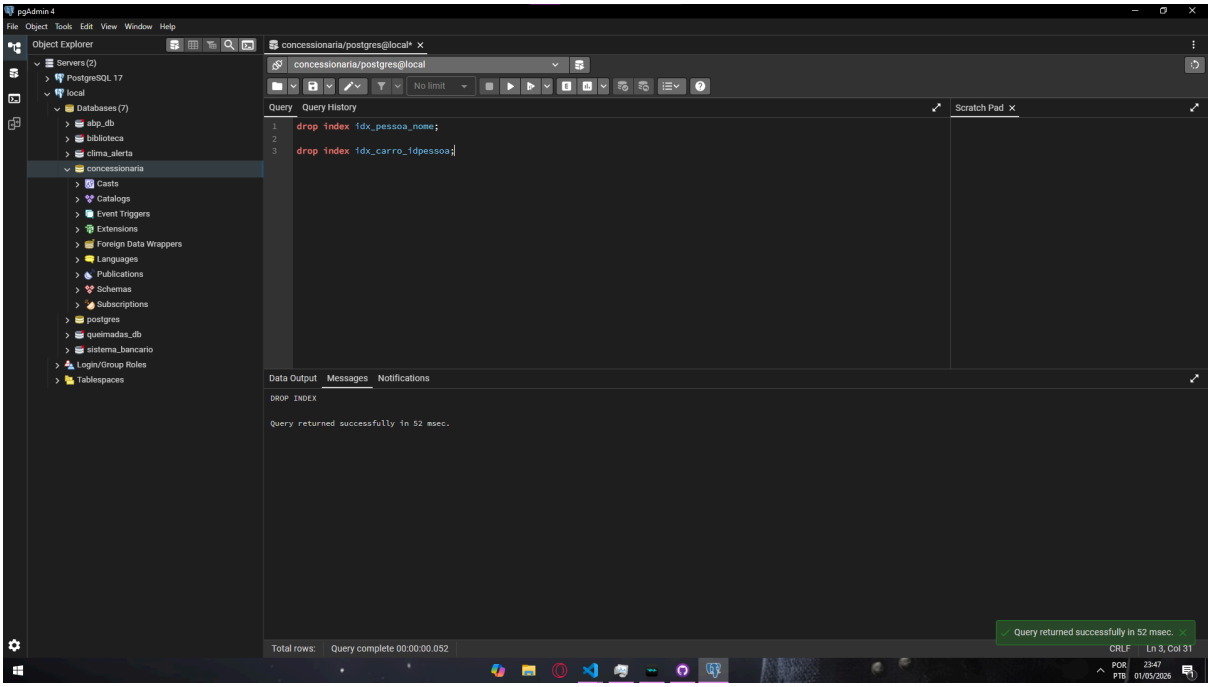


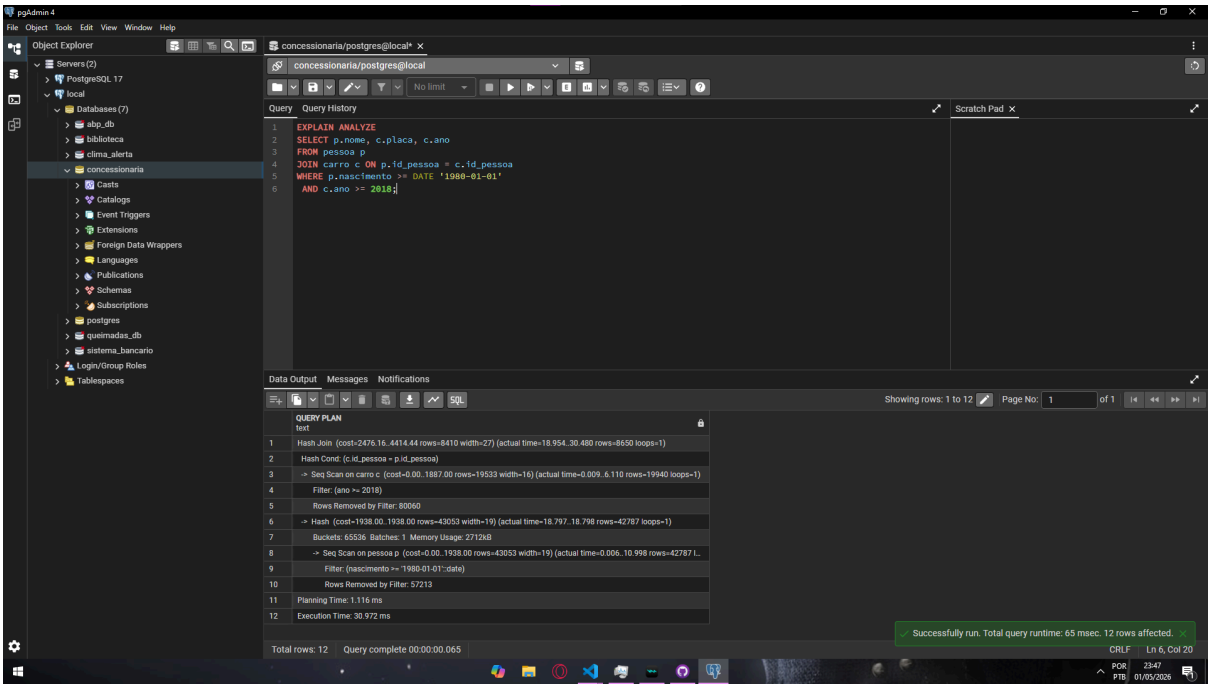


Sem índex: (mais lento)
Seq Scan

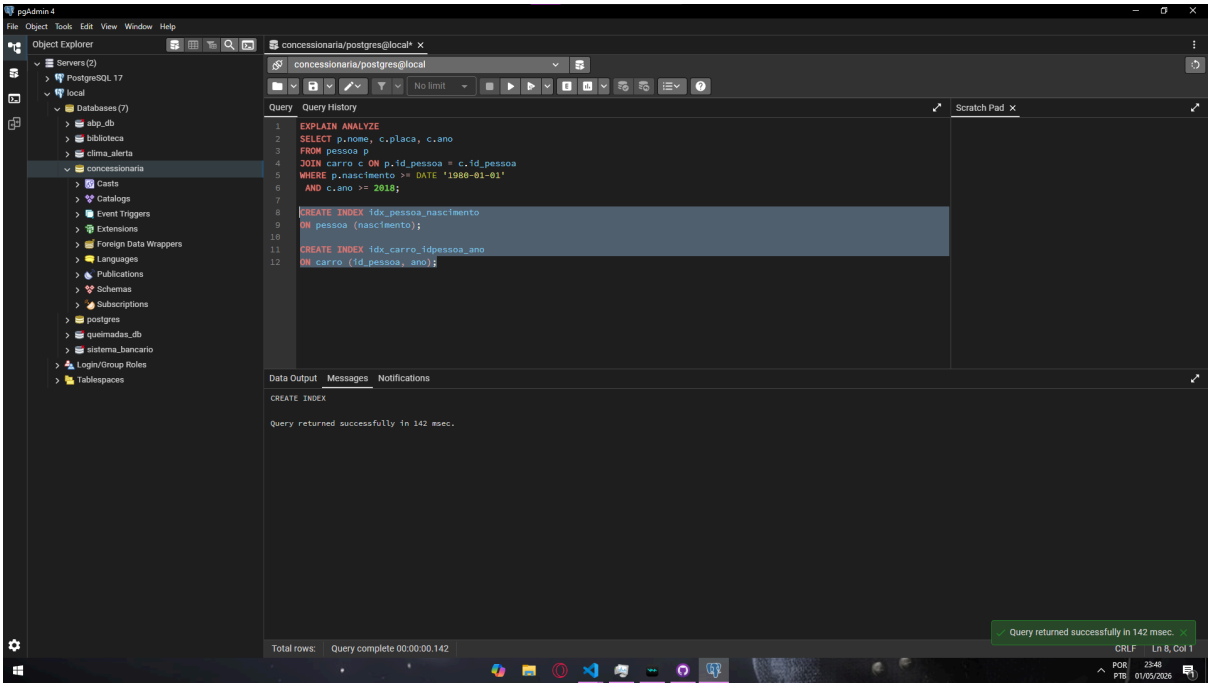
Com index: (mais rápido e join mais eficiente)
Bitmap Index Scan

7.



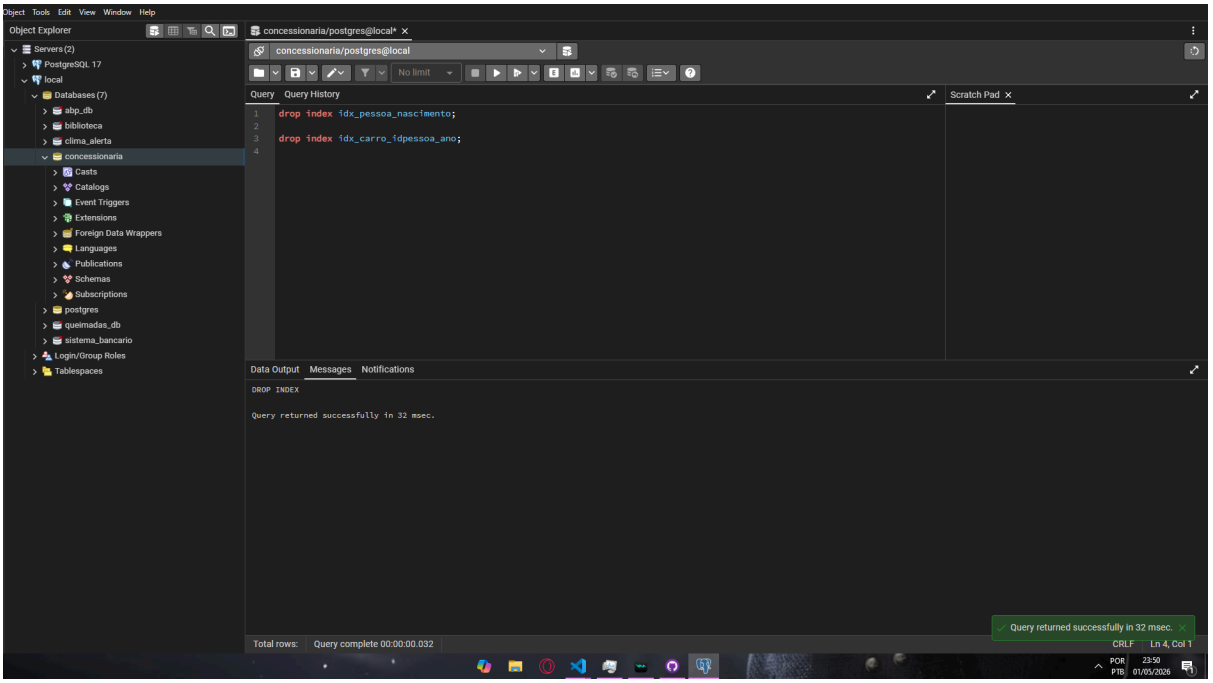


JOIN: p.id_pessoa = c.id_pessoa
Filtros: p.nascimento e c.ano

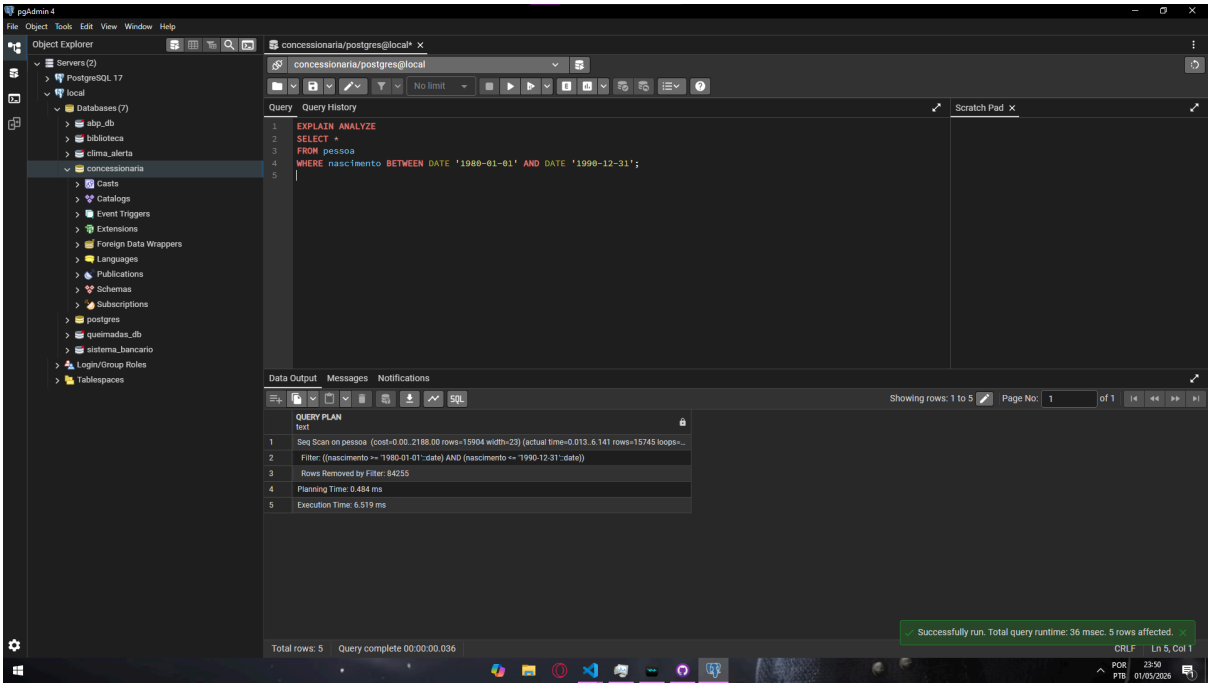


Índice em pessoa.nascimento para filtro
Índice composto em carro para JOIN e filtro

8.



A)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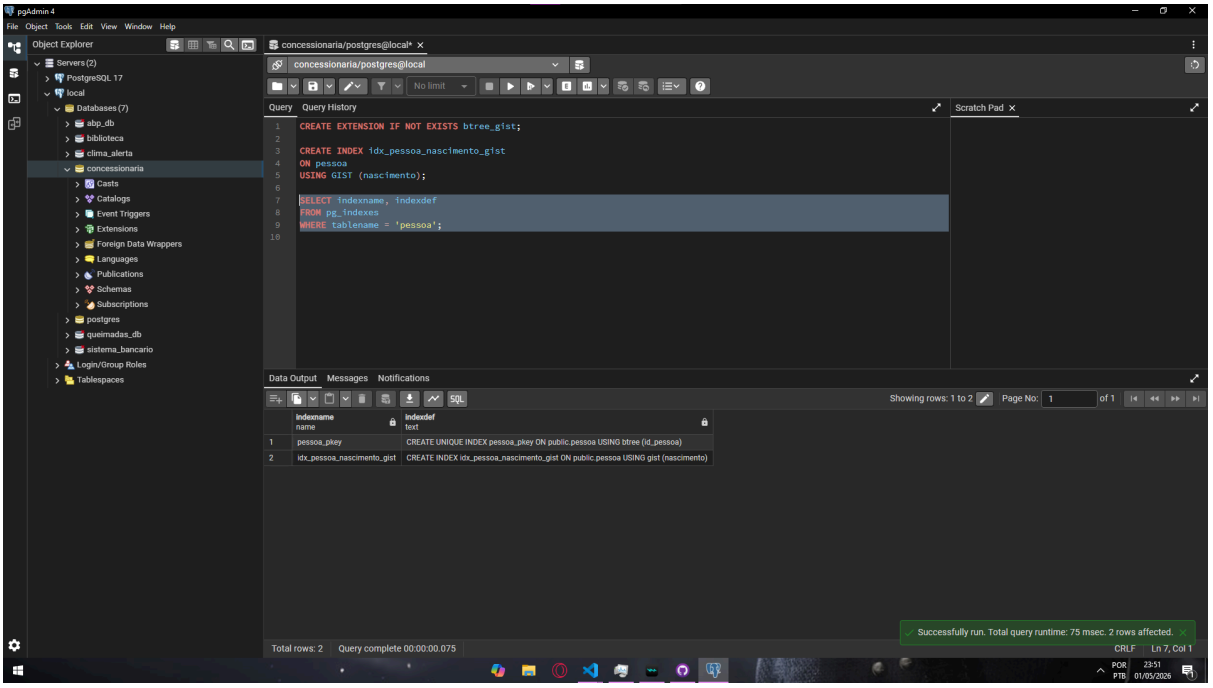
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
EXPLAIN ANALYZE
SELECT *
FROM pessoa
WHERE nascimento BETWEEN DATE '1980-01-01' AND DATE '1990-12-31';
Data Output Messages Notifications
Showing rows: 1 to 5
Page No: 1 of 1
QUERY PLAN
text
1 Seq Scan on pessoa (cost=0.00..2188.00 rows=15904 width=23) (actual time=0.013..6.141 rows=15745 loops=...
2 Filter: ((nascimento >= '1980-01-01'::date) AND (nascimento <= '1990-12-31'::date))
3 Rows Removed by Filter: 84255
4 Planning Time: 0.484 ms
5 Execution Time: 6.519 ms
Successfully run. Total query runtime: 36 msec. 5 rows affected.
Total rows: 5 Query complete 00:00:00.036
CRLF Ln 5, Col 1
POR PTB 23:50 01/05/2026

B)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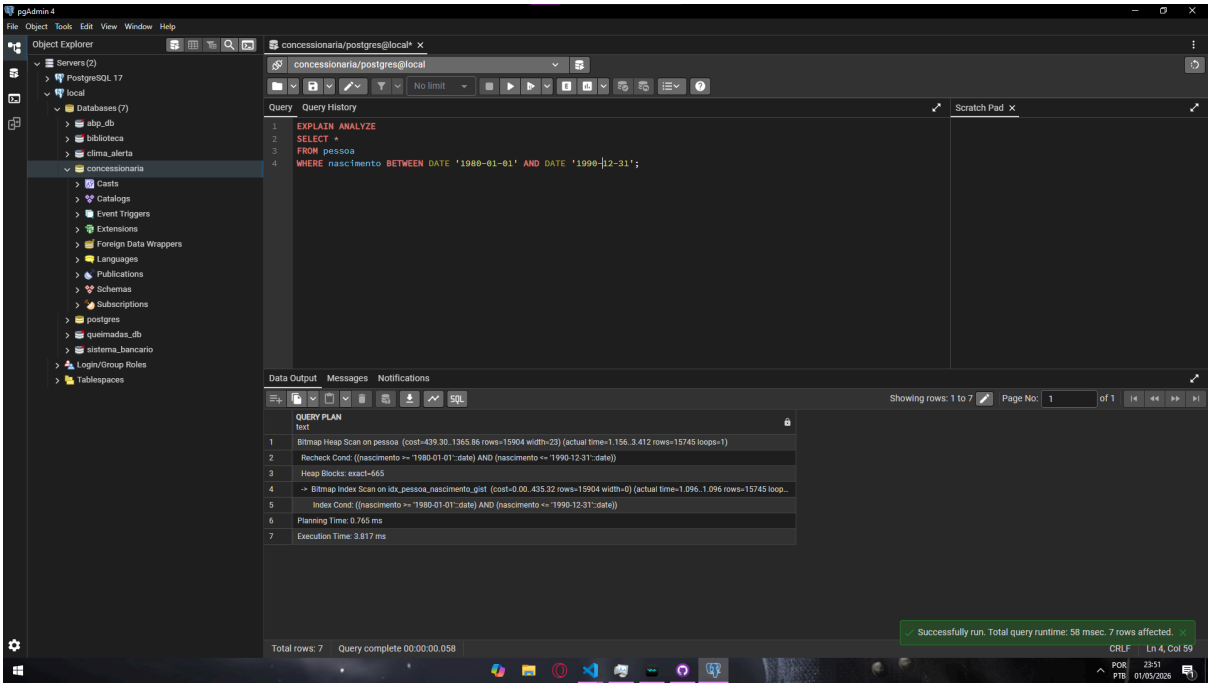
Login/Group Roles
Tablespaces
concessionaria/postgres@local*
concessionaria/postgres@local
No limit
Query Query History
Scratch Pad
CREATE EXTENSION IF NOT EXISTS btree_gist;
CREATE INDEX idx_pessoa_nascimento_gist
ON pessoa
USING GIST (nascimento);
SELECT indexname, indexdef
FROM pg_indexes
WHERE tablename = 'pessoa';
Data Output Messages Notifications
Showing rows: 1 to 2
Page No: 1 of 1
indexname
name
indexdef
text
1 pessoa_pkey CREATE UNIQUE INDEX pessoa_pkey ON public.pessoa USING btree (id_pessoa)
2 idx_pessoa_nascimento_gist CREATE INDEX idx_pessoa_nascimento_gist ON public.pessoa USING gist (nascimento)
Successfully run. Total query runtime: 75 msec. 2 rows affected.
Total rows: 2 Query complete 00:00:00.075
CRLF Ln 7, Col 1
POR PTB 23:51 01/05/2026

C)



D)

• Qual plano foi utilizado antes da criação do índice GiST?

Seq Scan

• O índice GiST foi utilizado após a criação?

Sim

• Houve melhora no tempo de execução?

Sim, de aproximadamente 3ms. Não tanto eficiente quanto as demais.